

$$\begin{aligned}
 1. \quad 23.4\% &= \frac{23.4}{100} \\
 &= \frac{234}{1000} \\
 &= \frac{117}{500}
 \end{aligned}$$

$$23.4\% = 0.234$$

$$\begin{aligned}
 \frac{3}{8} &= 0.375 \\
 &= 0.375 \times 100\% \\
 &= 37.5\%
 \end{aligned}$$

Fraction	Decimal	Percentage
$\frac{1}{4}$	0.25	25%
$\frac{117}{500}$	0.234	23.4%
$\frac{3}{8}$	0.375	37.5%

EXAM TIP

A percentage is a fraction with denominator 100, so 23.4%

$= \frac{23.4}{100}$. To change 23.4% into a decimal, divide 23.4 by 100.

To change a decimal into a percentage, multiply it by 100%.

$$2. \quad 8 : 200 = 1 : 25$$

$$2 \text{ cm} : 5 \text{ m} = 2 \text{ cm} : 500 \text{ cm} = 1 : 250$$

$$2 \text{ g} : 0.5 \text{ kg} = 2 \text{ g} : 500 \text{ g} = 1 : 250$$

$$0.4 : 10 = 4 : 100 = 1 : 25$$

The equivalent ratios are **8 : 200** and **0.4 : 10**.

EXAM TIP

Recall that 1 m = 100 cm and 1 kg = 1000 g.

$$\begin{aligned}
 3. \quad \text{Alyssa's salary at the start of 2022} &= \$76\,000 \left(1 + \frac{5.4}{100}\right)^2 \\
 &= \$84\,429.62 \text{ (to 2 d.p.)}
 \end{aligned}$$

EXAM TIP

The principal changes at the end of each year, so apply the

formula for compound interest, total amount $= P \left(1 + \frac{r}{100}\right)^2$.

$$4. \quad (\text{a}) \text{ Mode} = \text{Detective}$$

EXAM TIP

The mode refers to the category that occurs the most frequently. Do not write '15' as the answer.

$$\begin{aligned}
 (\text{b}) \text{ Total number of adults} &= 9 + 4 + 15 + 10 + 7 \\
 &= 45
 \end{aligned}$$

$$5. \quad (\text{a}) \quad 5, 2, -1, -4, -7, -10$$

EXAM TIP

Subtract 3 to get the next term.

$$(\text{b}) \text{ Common difference} = 7$$

$$n\text{th term} = 7n - 3$$

EXAM TIP

Identify the common difference, 7, between the consecutive terms of this sequence first.

$$\begin{aligned}
 6. \quad \angle DEA &= 180^\circ - 105^\circ \text{ (adj. } \angle \text{s on a str. line)} \\
 &= 75^\circ
 \end{aligned}$$

$$\begin{aligned}
 \text{Sum of angles in } ABCDE &= (5 - 2) \times 180^\circ \\
 &= 540^\circ
 \end{aligned}$$

$$93^\circ + \angle ABC + 134^\circ + 122^\circ + 75^\circ = 540^\circ$$

$$\angle ABC + 424^\circ = 540^\circ$$

$$\angle ABC = 116^\circ$$

$$\therefore \angle DEA = 75^\circ \text{ and } \angle ABC = 116^\circ$$

EXAM TIP

Sum of interior angles in an n -sided polygon $= (n - 2) \times 180^\circ$

$$7. \quad \text{Width of rectangle} = x \text{ cm}$$

$$\text{Length of rectangle} = 2x \text{ cm}$$

$$\text{Perimeter of rectangle} = 2(2x + x)$$

$$= 6x \text{ cm}$$

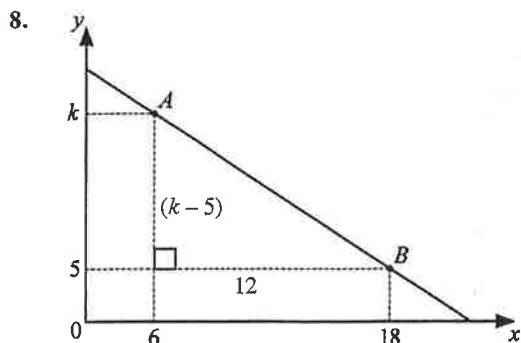
$$\text{Circumference of circle} = 2\pi(9)$$

$$= 18\pi \text{ cm}$$

$$\text{Perimeter of rectangle} = \text{Circumference of circle}$$

$$6x = 18\pi$$

$$x = 3\pi \text{ (shown)}$$



Using Pythagoras' Theorem,

$$(k - 5)^2 + 12^2 = 15^2$$

$$(k - 5)^2 = 15^2 - 12^2$$

$$= 81$$

$$k - 5 = 9 \text{ or } k - 5 = -9$$

$$k = 14$$

$$k = -4$$

Since $k > 5$, then $k = 14$.

EXAM TIP

Apply Pythagoras' Theorem to the right-angled triangle.

$$9. \text{ Cost price in 2022} = \frac{\$2.46}{102.5} \times 100$$

$$= \$2.40$$

EXAM TIP

Find 100% given that 102.5% represents \$2.46.

$$10. d = kt^2 - (1)$$

Substitute $d = 180$ and $t = 6$ into (1):

$$180 = k(6)^2$$

$$= 36k$$

$$k = 5$$

$$\therefore d = 5t^2$$

When $t = 3$,

$$d = 5(3)^2$$

$$= 45$$

$\therefore$ The ball falls **45 m** in the first 3 seconds.

EXAM TIP

Recall $y = kx$ if y is directly proportional to x , where k is a constant.

11. (a) In this case, the mode only considers the most frequently occurring value, which is the largest value in this set of data and is an outlier. It does not account for the distribution and spread of the data.

$$(b) \text{ Mean age} = \frac{20 + 21 + 22 + \dots + 46}{11}$$

$$= 31.8 \text{ years (to 3 s.f.)}$$

EXAM TIP

Apply mean = $\frac{\text{sum of the ages}}{\text{total number of players}}$.

$$12. (a) 125\,000 = 50^3$$

$$= (2 \times 5^2)^3$$

$$= 2^3 \times 5^6$$

$$\therefore a = 3, b = 6$$

EXAM TIP

Apply $(a^m)^n = a^{mn}$.

$$(b) (i) \text{ HCF} = 5^2 \times 7$$

$$(ii) \text{ LCM} = 2^3 \times 3^2 \times 5^3 \times 7^2$$

$$13. (a) \text{ Equation of line of symmetry: } x = \frac{-2+6}{2}$$

$$x = 2$$

EXAM TIP

The line of symmetry for a quadratic curve that cuts the x -axis at 2 distinct points is equidistant from the 2 x -intercepts.

$$(b) \text{ Equation of curve: } y = (x+2)(x-6) \text{ --- (1)}$$

When $x = 2$,

$$y = (2+2)(2-6)$$

$$= -16$$

$\therefore$ Coordinates of minimum point are **(2, -16)**.

$$14. \text{ Volume of water} = \frac{1}{3} \pi (8)^2 (20)$$

$$= \frac{1280\pi}{3} \text{ cm}^3$$

$$\text{Time taken} = \frac{1280\pi}{3} \div 200$$

$$= 6.70 \text{ s (to 3 s.f.)}$$

EXAM TIP

Recall volume of cone = $\frac{1}{3} \pi r^2 h$.

Time taken to fill the cone = Volume of cone $\div$ rate of filling the cone

$$15. (a) P + Q = 6.5 \times 10^n + 2.4 \times 10^n$$

$$= (6.5 + 2.4) \times 10^n$$

$$= 8.9 \times 10^n$$

$$(b) P \times Q = (6.5 \times 10^n) \times (2.4 \times 10^n)$$

$$= (6.5 \times 2.4) \times (10^n \times 10^n)$$

$$= 15.6 \times 10^{2n}$$

$$= 1.56 \times 10^{2n+1}$$

EXAM TIP

Apply $a^m \times a^n = a^{m+n}$ or $(a^m)^n = a^{mn}$.

$$16. \text{ Area of shaded segment} = \text{Area of sector} - \text{area of triangle}$$

$$= \frac{65^\circ}{360^\circ} \times \pi (7)^2 - \frac{1}{2} (7)(7) \sin 65^\circ$$

$$= 5.59 \text{ cm}^2 \text{ (to 3 s.f.)}$$

EXAM TIP

Recall area of sector = $\frac{\theta^\circ}{360^\circ} \times \pi r^2$ and area of triangle = $\frac{1}{2} ab \sin C$.

$$17. (a) 3x^2 + 2x - 8 = (3x - 4)(x + 2)$$

EXAM TIP

Use a multiplication frame to factorise the quadratic expression.

$$(b) \frac{5}{x^2 - 16} - \frac{1}{x + 4} = \frac{5}{(x+4)(x-4)} - \frac{1}{x+4}$$

$$= \frac{5 - (x-4)}{(x+4)(x-4)}$$

$$= \frac{5 - x + 4}{(x+4)(x-4)}$$

$$= \frac{9 - x}{(x+4)(x-4)}$$

EXAM TIP

Recall the special algebraic identity: $a^2 - b^2 = (a+b)(a-b)$.

18. (a) $\angle DCE = 40^\circ$ (vert. opp. $\angle$ s)
 $\angle CDE = 180^\circ - 40^\circ - 67^\circ$ ($\angle$ sum of $\triangle$)
 $= 73^\circ$

(b) Since $\triangle ABC$ is similar to $\triangle EDC$,

$$\frac{CE}{CA} = \frac{DE}{BA}$$

$$= \frac{9}{6}$$

$$= 1.5$$

$$CE = 1.5CA$$

Given that $AC + CE = 22.3$ cm,

$$AC + 1.5AC = 22.3$$

$$2.5AC = 22.3$$

$$AC = 8.92$$
 cm

$$\therefore CE = 1.5(8.92)$$

$$= 13.38$$
 cm

EXAM TIP

Since the triangles are similar, their corresponding sides are in the same ratio. Use the fact that $AC + EC = AE$ to find the length of CE .

19. $7x + 3y = 9$ — (1)
 $3x - 5y = 7$ — (2)
 $(1) \times 5: 35x + 15y = 45$ — (3)
 $(2) \times 3: 9x - 15y = 21$ — (4)
 $(3) + (4): 44x = 66$
 $x = 1.5$

Substitute $x = 1.5$ into (1):

$$7(1.5) + 3y = 9$$

$$10.5 + 3y = 9$$

$$3y = -1.5$$

$$y = -0.5$$

$$\therefore x = 1.5, y = -0.5$$

EXAM TIP

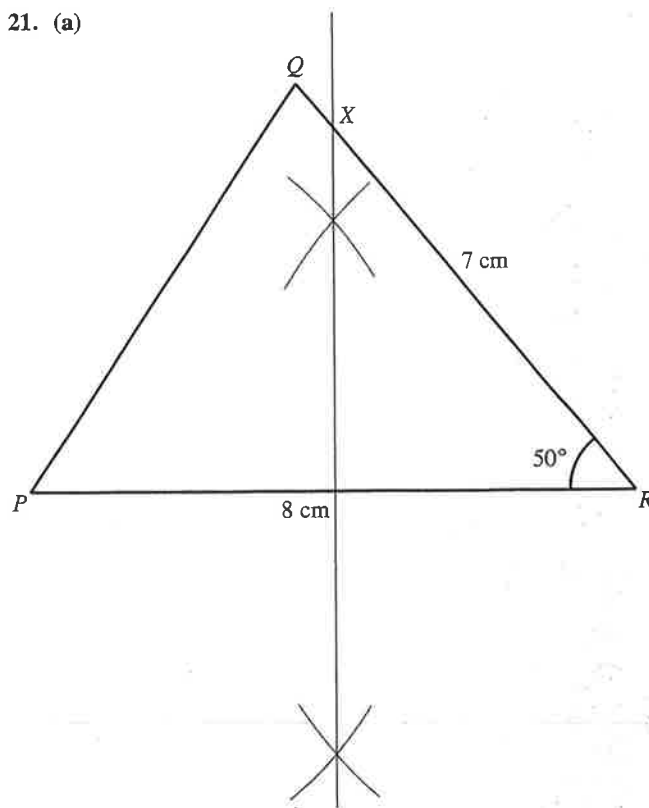
Apply the elimination or substitution method.

20. $\frac{6}{x-2} = x-1$
 $6 = (x-2)(x-1)$
 $= x^2 - 3x + 2$
 $x^2 - 3x - 4 = 0$
 $(x-4)(x+1) = 0$
 $x-4 = 0$ or $x+1 = 0$
 $x = 4$ or $x = -1$
 $\therefore x = 4$ or $x = -1$

EXAM TIP

Multiply both sides of the equation by $x-2$. Then use a multiplication frame to factorise the quadratic expression $x^2 - 3x - 4$.

21. (a)



- (b) $\angle QPR = 57^\circ$

- (c) $PX = 6.2$ cm

$$\text{Actual length of } PX = 6.2 \times 0.5$$

$$= 3.1 \text{ m}$$

22. (a)

		First dice					
		1	2	3	4	5	6
Second dice	1	0	1	2	3	4	5
	2	1	0	1	2	3	4
	3	2	1	0	1	2	3
	4	3	2	1	0	1	2
	5	4	3	2	1	0	1
	6	5	4	3	2	1	0

- (b) A difference of 5 occurs 2 times out of the 36 possible outcomes, so $\frac{2}{36} = \frac{1}{18}$.

- (c) $P(\text{difference} < 2) = \frac{16}{36}$
 $= \frac{4}{9}$

EXAM TIP

Count the outcomes with a difference of 0 and 1 only.

23. (a) Width = x cm
 Height = $(x - 4)$ cm
 Length = $3x$ cm
 Surface area = $2[x(x - 4) + x(3x) + (x - 4)(3x)]$
 $= 2(x^2 - 4x + 3x^2 + 3x^2 - 12x)$
 $= 2(7x^2 - 16x)$
 $= 14x^2 - 32x$ (shown)

EXAM TIP

Total surface area of cuboid = Area of 6 rectangular faces

- (b) Given that surface area = 2670 cm^2 ,

$$\begin{aligned} 14x^2 - 32x &= 2670 \\ 14x^2 - 32x - 2670 &= 0 \\ 7x^2 - 16x - 1335 &= 0 \\ x &= \frac{-(-16) \pm \sqrt{(-16)^2 - 4(7)(-1335)}}{2(7)} \\ &= \frac{16 \pm \sqrt{37\ 636}}{14} \\ &= \frac{16 \pm 194}{14} \\ &= 15 \text{ or } -12\frac{5}{7} \end{aligned}$$

$\therefore$ Width of cuboid = **15 cm**

EXAM TIP

Alternatively, factorise $7x^2 - 16x - 1335$ using a multiplication frame to obtain $(7x + 89)(x - 15)$.

- (b) $\angle XGF = 90^\circ$
 $\angle GFE = 135^\circ$
 $\therefore \angle XEF = 360^\circ - 60^\circ - 90^\circ - 135^\circ$ ($\angle$ sum of a quad.)
 $= 75^\circ$

4. (a) $40 < m \leq 50$

EXAM TIP

$\frac{100}{2} = 50$, so the median mass is the mean of the values in the 50th and the 51st positions.

- (b) Estimated mean

$$\begin{aligned} &= \frac{12(35) + 40(45) + 25(55) + 16(65) + 7(75)}{100} \\ &= 51.6 \text{ kg} \end{aligned}$$

EXAM TIP

Apply mean = $\frac{\sum fx}{\sum f}$, where f is the frequency and x is the mid-value of each interval.

- (c) (i) Angle of sector = $\frac{16}{100} \times 360^\circ$
 $= 57.6^\circ$

- (ii) The frequencies of the categories $30 < m \leq 40$ and $60 < m \leq 70$ are quite close to each other, so the size of the sectors of a pie chart might not be able to show the contrast between these frequencies as clearly as the heights of the bars of a histogram.

5. (a) $5x^2y(3x^3y^2 + 4x^2y) = 15x^5y^3 + 20x^4y^2$

EXAM TIP

Apply $a^m \times a^n = a^{m+n}$.

- (b) $(2a - 3b)^2 = (2a)^2 - 2(2a)(3b) + (3b)^2$
 $= 4a^2 - 12ab + 9b^2$

- (c) $P = 5Q - 2R^2$
 $2R^2 = 5Q - P$
 $R^2 = \frac{5Q - P}{2}$
 $R = \pm \sqrt{\frac{5Q - P}{2}}$

EXAM TIP

To make R the subject of the formula, rearrange the formula such that R (or R^2 first) is alone on the LHS of the equation.

- (d) $\frac{x+5}{3} - \frac{x-2}{5} = 7$
 $5(x+5) - 3(x-2) = 105$
 $5x + 25 - 3x + 6 = 105$
 $2x = 74$
 $x = 37$

EXAM TIP

Multiply both sides of the equation by 15.

1. (a) $270\ 698 = 270\ 700$ (to 4 s.f.)
 (b) $\frac{36.72 \times 224.3}{289.3 - 210.7} \approx \frac{40 \times 200.3}{300 - 200}$
 $= \frac{8000}{100}$
 $= 80$
2. (a) $96 \text{ km/h} = \frac{(96 \times 1000) \text{ m}}{(1 \times 3600) \text{ s}}$
 $= 26\frac{2}{3} \text{ m/s}$
 (b) Time taken = $\frac{240 + 180}{26\frac{2}{3}}$
 $= 15.75 \text{ s}$

EXAM TIP

Include the length of the train in the total distance.

3. (a) Interior angle of regular octagon = $\frac{(8-2) \times 180^\circ}{8}$
 $= 135^\circ$

EXAM TIP

Sum of interior angles in an n -sided polygon = $(n-2) \times 180^\circ$

6. (a) 1 cm represents 50 000 cm = 0.5 km.

$$\begin{aligned}\text{Length of road on the map} &= \frac{4}{0.5} \text{ cm} \\ &= 8 \text{ cm}\end{aligned}$$

EXAM TIP

1 : 50 000 means 1 cm on the map represents 50 000 cm on the actual ground. Recall that 1 km = 1000 m and 1 m = 100 cm, so 1 km = 100 000 cm.

- (b) (i) Initial lengths: $5x$ cm and $3x$ cm

New lengths: $(5x - 6)$ cm and $(3x - 6)$ cm

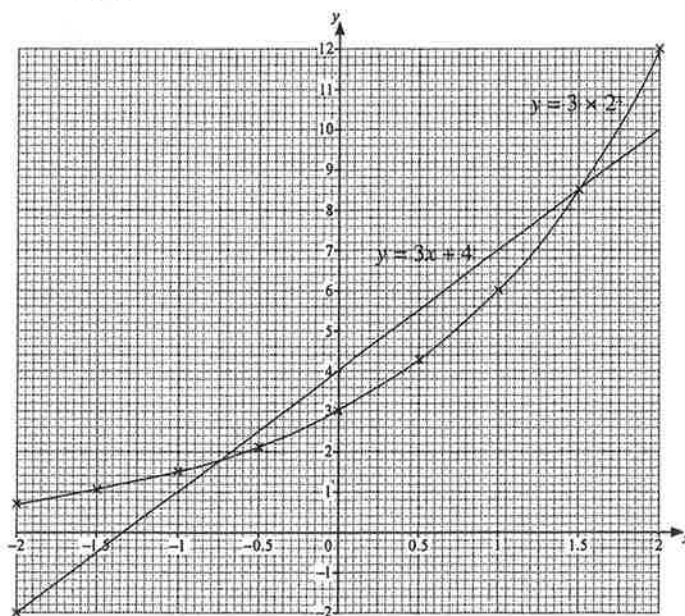
$$\begin{aligned}\frac{5x-6}{3x-6} &= \frac{9}{5} \\ 5(5x-6) &= 9(3x-6) \\ 25x-30 &= 27x-54 \\ 2x &= 24 \\ x &= 12\end{aligned}$$

- (ii) Original length of the longer piece of string = $5(12)$
= 60 cm

7. (a)

x	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2
y	0.75	1.06	1.5	2.12	3	4.24	6	8.49	12

- (b), (c)

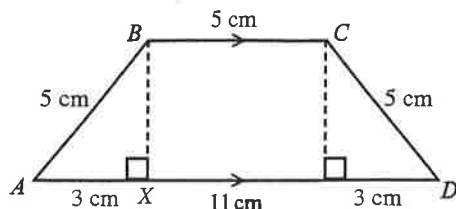


- (d) From the graph, x -coordinates of the points of intersection are -0.75 and 1.5 .

8. (a) $AB + BC + CD + 11 = 26$

$$3AB = 15$$

$$AB = 5 \text{ cm}$$



Using Pythagoras' Theorem,

$$BX^2 + 3^2 = 5^2$$

$$BX^2 = 5^2 - 3^2$$

$$= 16$$

$$BX = \sqrt{16} \quad (BX > 0)$$

$$= 4 \text{ cm}$$

$$\begin{aligned}\therefore \text{Area of trapezium} &= \frac{1}{2} \times (11 + 5) \times 4 \\ &= 32 \text{ cm}^2\end{aligned}$$

EXAM TIP

Recall area of trapezium = $\frac{1}{2} \times \text{sum of parallel sides} \times h$.

(b) $\sin \angle ABX = \frac{3}{5}$

$$\angle ABX = \sin^{-1} \frac{3}{5}$$

$$= 36.870^\circ \text{ (to 3 d.p.)}$$

$$\therefore \angle ABC = 36.870^\circ + 90^\circ$$

$$= 126.9^\circ \text{ (to 1 d.p.)}$$

EXAM TIP

Since $\triangle ABX$ is a right-angled triangle, apply

$$\sin x^\circ = \frac{\text{opposite side}}{\text{hypotenuse}}$$

9. (a) Gradient of $AB = \frac{7-3}{2-8}$
 $= -\frac{2}{3}$

Substitute $x = 2, y = 7, m = -\frac{2}{3}$ into $y = mx + c$:

$$7 = -\frac{2}{3}(2) + c$$

$$= -\frac{4}{3} + c$$

$$c = \frac{25}{3}$$

$$\therefore \text{Equation of } AB: y = -\frac{2}{3}x + \frac{25}{3}$$

$$3y = 25 - 2x$$

EXAM TIP

Express the equation of the graph in the form $y = mx + c$.

(b) x -coordinate of $M = \frac{2+8}{2} = 5$

$$y\text{-coordinate of } M = \frac{7+3}{2} = 5$$

Substitute $y = 5$ into LHS of $2y = 3x - 5$:

$$2y = 2(5) = 10$$

Substitute $x = 5$ into RHS of $2y = 3x - 5$:

$$3x - 5 = 3(5) - 5 = 10$$

Since $x = 5, y = 5$ satisfies the equation, L passes through M .

$\therefore$ Asif is correct.

EXAM TIP

If the coordinates of M satisfy the equation L , then M lies on L .

$$(c) \quad 2y = 3x - 5$$

$$y = \frac{3}{2}x + \frac{5}{2}$$

$$\text{Gradient of } L = \frac{3}{2}$$

$$\text{Product of gradients} = -\frac{2}{3} \times \frac{3}{2}$$

$$= -1$$

$\therefore$ The line AB is perpendicular to the line L .

EXAM TIP

Recall the gradient of a line with equation $y = mx + c$ is m .

$$10. (a) \quad \text{Total number of stars} = 4 \times 2.5$$

$$= 10$$

$$\text{Number of stars solar energy scored for Efficiency}$$

$$= 10 - 1 - 3 - 3$$

$$= 3$$

EXAM TIP

The mean is 4 and there are 4 categories, so calculate the total number of stars first.

(b) *Graph in the question has been removed due to copyright restrictions.*

$$(c) (i) \quad \text{Amount of electricity used by the solar water heater}$$

$$= 365 \times 4 \times 3000 \times 50\% \text{ watt-hours}$$

$$= 2\,190\,000 \text{ watt-hours}$$

$$= 2190 \text{ kilowatt-hours (shown)}$$

$$(ii) \quad \text{Amount of electricity used by the heat pump water heater}$$

$$= 365 \times 4 \times (200 + 3000 \times 2\%) \text{ watt-hours}$$

$$= 379\,600 \text{ watt-hours}$$

$$= 379.6 \text{ kilowatt-hours}$$

$$\text{Difference in electricity used}$$

$$= (2190 - 379.6) \text{ kilowatt-hours}$$

$$= 1810.4 \text{ kilowatt-hours}$$

$$\approx 1800 \text{ kilowatt-hours}$$

$$\therefore \text{Huan is correct.}$$

$$11. (a) \quad \text{Angle } BCD = 1.94 \text{ rad because angles in opposite segments, i.e. } \angle BAD \text{ and } \angle BCD, \text{ are supplementary.}$$

EXAM TIP

Recall that angles in opposite segments are supplementary, i.e. they have a sum of π .

(b) Using Cosine Rule,

$$BD^2 = AB^2 + AD^2 - 2(AB)(AD) \cos \angle BAD$$

$$= 3.6^2 + 5.2^2 - 2(3.6)(5.2) \cos 68.8^\circ$$

$$= 26.461 \text{ (to 5 s.f.)}$$

$$BD = \sqrt{26.461} \text{ (} BD > 0 \text{)}$$

$$= 5.14 \text{ cm (to 3 s.f.)}$$

EXAM TIP

The Cosine Rule relates three sides and one angle.

(c) Using Sine Rule,

$$\frac{\sin \angle CBD}{CD} = \frac{\sin \angle BCD}{BD}$$

$$\frac{\sin \angle CBD}{2.1} = \frac{\sin 111.2^\circ}{5.1440}$$

$$\sin \angle CBD = \frac{2.1 \sin 111.2^\circ}{5.1440}$$

$$= 0.380\,61 \text{ (to 5 s.f.)}$$

$$\angle CBD = \sin^{-1} 0.380\,61$$

$$= 22.372^\circ \text{ (to 3 d.p.)}$$

$$\therefore \angle BDC = 180^\circ - 111.2^\circ - 22.372^\circ \text{ (} \angle \text{ sum of } \triangle \text{)}$$

$$= 46.4^\circ \text{ (to 1 d.p.)}$$

EXAM TIP

The Sine Rule, which relates two sides and two angles, may be used to calculate $\angle CBD$, before using the property of the angle sum of a triangle to calculate $\angle BDC$.

12. (a) (i) Variety A has a smaller standard deviation, so the total mass of potatoes from the 5 plants will have a smaller spread.

(ii) Variety B has a greater mean, indicating that on average, the total mass of potatoes from the 50 plants will be greater than if Feng were to choose variety A .

(b) (i) Median = 29 cm

(ii) Upper quartile = 38 cm

Lower quartile = 21 cm

$\therefore$ Interquartile range

$$= (38 - 21) \text{ cm}$$

$$= 17 \text{ cm}$$

EXAM TIP

Recall that interquartile range = upper quartile - lower quartile.

(iii) P(one has a height of less than 10 cm and the other has a height of more than 40 cm)

$$= \frac{6}{120} \times \frac{24}{119} + \frac{24}{120} \times \frac{6}{119}$$

$$= \frac{12}{595}$$